

MD HASIBUL AMIN

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LinkedIn

Google Scholar

EDUCATION

- **University of South Carolina** *Jan 2021-present*
PhD in Computer Engineering
Research topic: Processing-in-Memory (PIM), ML System Optimization, ML software-hardware co-design, ML/AI Hardware Accelerators
- **University of South Carolina** *Jan 2021-May 2024*
MS in Computer Engineering
Selected Courseworks: Advanced Digital Design, Computer Architecture, Pattern Recognition, Robotics, Analysis of Algorithms
- **Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh** *Feb 2015-Apr 2019*
Bachelor of Science in Electrical and Electronic Engineering (EEE)
Research topic: Ab initio study of Carbon Nanotubes
Thesis title: Ab initio investigation of dopants for ultrahigh conductivities in single wall carbon nanotubes

RESEARCH EXPERIENCE

- **In-memory Analog Computing (IMAC)** | **Advisor: Dr. Ramtin Zand** *Jan 2021-present*
 - Developed neural architecture search framework for PIM systems based on weight sharing and mixed-precision quantization along with evolutionary search [\[paper\]](#) [\[code\]](#)
 - Optimized crossbar circuit performance by modeling interconnect parasitics and analog partitioning [\[paper\]](#)
 - Built a Python-based SPICE simulation framework for in-memory accelerator circuit analysis [\[paper\]](#)
 - Designed novel analog activation function using emerging memory devices; Verilog-A device model development for circuit analysis in SPICE [\[paper\]](#)
- **Heterogeneous Architectures for AI** | **Advisor: Dr. Ramtin Zand** *Jan 2023-present*
 - Designed and implemented a hybrid TPU-PIM architecture for 1-bit LLMs, improving computational efficiency and reducing memory bottlenecks [\[paper\]](#)
 - Designed and analyzed a heterogeneous TPU-IMAC architecture for CNN models [\[paper\]](#)
- **Approximate TPU** | **Advisor: Dr. Ramtin Zand** *Jun 2021-Jan 2025*
 - Implemented Verilog RTL and HLS designs for approximate TPU circuits; automated benchmarking with Python and TCL scripts
 - Developed python and TCL scripts for automatic simulation and analysis of various approximate arithmetic circuits in Design Compiler, Vivado and Vitis HLS [\[paper\]](#)
- **Investigation of Dopants for Carbon Nanotubes** | **Advisor: Dr. Ahmed Zubair** *May 2018-Dec 2019*
 - Modeled metallic (6,6) SWCNT and added various dopants for ab initio simulation on Quantum Espresso software environment.
 - Investigated the effect of various dopants such as I_2 , $AuCl_3$ on various carbon nanotube molecules using Density functional theory (DFT) [\[paper\]](#)

PROFESSIONAL EXPERIENCE

- **Graduate Research/Teaching Assistant** *Jan 2021-present*
Department of Computer Science and Engineering, University of South Carolina
- **Lecturer, Department of Electrical and Electronic Engineering** *Jan 2020-May 2020*
Daffodil International University, Dhaka, Bangladesh

TECHNICAL SKILLS

- **Hardware Design:** Verilog, SystemVerilog, HLS
- **Architectural Simulation:** gem5, McPAT, CACTI, MNSIM, NeuroSim
- **Programming Languages:** C, C++, Java, Python, MATLAB, TCL
- **EDA Tools:** Cadence Virtuoso, Synopsys Design Compiler, HSPICE
- **Machine Learning:** PyTorch, TensorFlow, Quantization Techniques

PRESENTATIONS AND AWARDS

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| • Best Poster Award at GLSVLSI 2025 | <i>June 2025</i> |
| • Presenter at DAC PhD Forum 2025 | <i>June 2025</i> |
| • 2025 CSE Graduate Research Symposium Best Poster Award - 2nd Place | <i>March 2025</i> |
| • IEEE ISCAS 2022 Student Participation Grant | <i>May 2022</i> |

SELECTED ACADEMIC PROJECTS

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|--|--------------------|
| • Training a transformer-based language model for novel antimicrobial peptide generation | <i>Spring 2024</i> |
| • RTL design and FPGA implementation of a 3-stage pipelined RISC-V microarchitecture | <i>Fall 2021</i> |
| • RTL design of a 32-bit single-cycle MIPS microarchitecture | <i>Fall 2021</i> |
| • Circuit design for a Leaky Integrate-and-Fire (LIF) spiking neuron | <i>Spring 2021</i> |

PEER REVIEW SERVICE

- | | |
|---|-------------|
| • ISCAS-2025, ISVLSI-2025, TCAD | <i>2025</i> |
| • ISVLSI-2024, GLSVLSI-2024 | <i>2024</i> |
| • ISCAS-2023, GLSVLSI-2023, MWSCAS-2023, JETCAS | <i>2023</i> |

SELECTED PUBLICATIONS

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| • <u>M. H. Amin</u> , M. Mohammadi, J. D. Bakos and R. Zand. CrossNAS: A Cross-Layer Neural Architecture Search Framework for PIM Systems. GLSVLSI '25. (Best Poster Award) | <i>2025</i> |
| • S. V. Singh, A. Kolinko, <u>M. H. Amin</u> , R. Zand and J. D. Bakos. A Decomposition-Based Memristive Crossbar Solver and FPGA-Accelerated Hardware Implementation. GLSVLSI '25. | <i>2025</i> |
| • S. Tabrizchi, A. S. Sarvestani, <u>M. H. Amin</u> , D. Najafi, S. Angizi, R. Zand, A. Roohi. Magnetic In/Near-Sensor Architectures: From Raw Sensing to Smart Processing. GLSVLSI '25. | <i>2025</i> |
| • D. Vungarala, <u>M. H. Amin</u> , A. Roohi, A. Ghosh, R. Zand, S. Angizi. From Prompt to Accelerator: A Perspective on LLM-Based Analog In-Memory Accelerator Design Automation. GLSVLSI '25. | <i>2025</i> |
| • D. Vungarala, <u>M. H. Amin</u> , P. Mercati, A. Roohi, R. Zand, S. Angizi. LLM-IMC: Automating Analog In-Memory Computing Architecture Generation with Large Language Models. FCCM '25. | <i>2025</i> |
| • M. E. Elbtity, <u>M. H. Amin</u> , H. Hassan, et al. Design Automation and Quantitative Analysis of Approximate Arithmetic Circuits. TechRxiv. | <i>2024</i> |
| • <u>M. H. Amin</u> , M. E. Elbtity, and R. Zand. IMAC-Sim: A Circuit-level Simulator For In-Memory Analog Computing Architectures. In Proceedings of the Great Lakes Symposium on VLSI 2023 (GLSVLSI '23) | <i>2023</i> |
| • M. E. Elbtity, B. Reidy, <u>M. H. Amin</u> , and R. Zand. Heterogeneous Integration of In-Memory Analog Computing Architectures with Tensor Processing Units. GLSVLSI '23. | <i>2023</i> |
| • <u>M. H. Amin</u> , M. E. Elbtity and R. Zand. Xbar-Partitioning: A Practical Way for Parasitics and Noise Tolerance in Analog IMC Circuits. IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS) | <i>2022</i> |
| • <u>M. H. Amin</u> , M. Elbtity, M. Mohammadi, R. Zand. MRAM-based Analog Sigmoid Function for In-memory Computing. GLSVLSI '22. | <i>2022</i> |
| • <u>M. H. Amin</u> , M. Elbtity and R. Zand. Interconnect Parasitics and Partitioning in Fully-Analog In-Memory Computing Architectures. IEEE International Symposium on Circuits and Systems (ISCAS). | <i>2022</i> |
| • M. L. Rahman, <u>M. H. Amin</u> and A. Zubair. Ab initio Theoretical Investigation of Dopants for Ultrahigh Conductivities in Single Wall Carbon Nanotubes. IEEE Region 10 Conference (TENCON). | <i>2019</i> |